WASP-12b

R-0285

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-12_240228` · created 2026-07-17T16:48:06+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460367.7174 +/- 0.0012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.154 +/- 0.044
Transit depth (Rp/Rs)^2	2.38 +/- 1.36 [%]
Orbital Inclination (inc)	86.48 +/- 2.93
Airmass coefficient 1 (a1)	0.77 +/- 0.45
Airmass coefficient 2 (a2)	0.272 +/- 0.098
Scatter in the residuals of the lightcurve fit is	64.33 %
Best Comparison Star	None
Optimal Aperture	4.73
Optimal Annulus	11.82
Transit Duration (day)	0.1324 +/- 0.0066

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.154 ± 0.044 vs 0.1178 (deviation 0.77σ)
Duration fit vs published	0.1324 d vs 0.125 d (deviation 1.04σ)
Mid-time O–C	-1570.1 min
Depth SNR	3.3
Residual scatter	64.33 %

Light curve

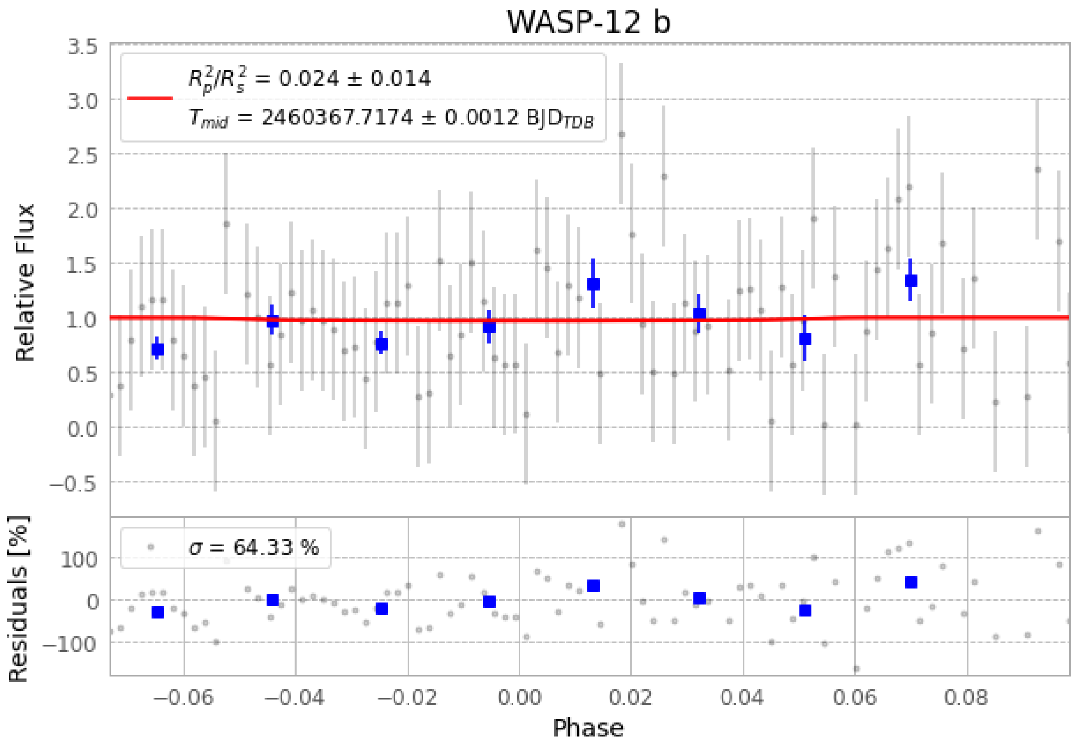

Figure 1 — FinalLightCurve_WASP-12 b_2024-02-26.png (SHA-256 39863581a672ff43…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [217, 361], 2 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 b22cda5dbd4c3e94…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit faa81f0`

Data provenance

91 FITS frames (60 MB), `WASP-12240227031215.FITS` → `WASP-12240227074215.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-240228.log (1f09a6538a08…), FinalLightCurve_WASP-12 b_2024-02-26.png (39863581a672…), FinalLightCurve_WASP-12 b_2024-02-26.csv (908ab9d10de1…)

Compute resources

Wall time 205.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)